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The Extended Gate Family

Six new handles beyond the protocol; the stability of the H2 gap

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Introduction and setting

Campaign C5 answers the natural objection to C3: perhaps identity is reached not on the protocol handles Nv, q, W, α but on something else? We extend the handle space by six new dimensions: boundary twists (φx, φy) — spin structures in the sense of C8; hopping anisotropy t_x/t_y ∈ {0.8, 1.25}; flux shift δα = ±0.02 about the self-dual line (±1/24 on the torus, where the constraint α·Ny ∈ ℤ is exact); a staggered mass m ∈ {0.15, 0.35} breaking the chiral structure; the vortex profile — point-like (site) versus a finite-radius core (midpoint); the charge sign pattern — neutral (protocol), all-plus, random; and the boundary type — open versus torus.

In total 35 new gates plus 2 random-sign decoys, all at the same master seed 96, the same unfolding pipeline and the same three statistics as C3 — the comparison is batch-consistent.

Method and safeguards

Every handle is implemented in the engine.hamiltonian kernel and covered by a regression control: the protocol reference gate (Nv = 0, W = 0.5, α = 0.5) reproduces the mean level spacing of the C3 CSV bit-for-bit (deviation 1.4e-16) — the extended pipeline has not shifted the base. The degenerate corner W = 0 with the midpoint profile produces exact multiplicities and breaks the polynomial unfolding; that corner is replaced by W = 0.1 with an explicit note in the data table.

On the torus the flux shift is rounded to the realisable ±1/24: the constraint α·Ny ∈ ℤ on the 24-lattice is rigid, and these are exact neighbouring points of the self-dual line, not approximations.

Results: the gap is stable

The main result is negative and therefore valuable. All 35 new gates sit in the GUE class: ⟨r⟩ = 0.498–0.505 (0.500 ± 0.002 across the family). The correlation with the zeros satisfies |r_H7| ≤ 0.043 against the best protocol value 0.226 and positive controls of 1.000; the trace projection r_EF (C4) is distributed as 0.52 ± 0.52 — exactly inside the null clouds (surrogate, Poisson, GUE), and the decoys behave the same. No handle — alone or in combinations (anisotropy+mass, twist+mass, midpoint+torus) — moves a gate out of the null zone.

Handle by handle: twists give |r_H7| ≤ 0.033 (spin structures change the global phase but not the spectral class); anisotropy ≤ 0.030; flux shift ≤ 0.034; mass ≤ 0.032 — the chiral breaking is the safest; profile ≤ 0.033; signs and boundaries ≤ 0.044. The family maximum is reached on a random-sign decoy — the expected noise maximum.

Table 1. The six new handles: maxima of |r_H7|

Handle	Values	max |r_H7|
Twists (φx, φy)	3 spin-structure classes	0.033
Anisotropy t_x/t_y	0.8; 1.25	0.030
Flux shift δα	±0.02; ±1/24 (torus)	0.034

Handle	Values	max |r_H7|
Staggered mass m	0.15; 0.35	0.032
Vortex profile	site; midpoint	0.033
Signs and boundaries	neutral/all_plus/random; open/torus	0.044

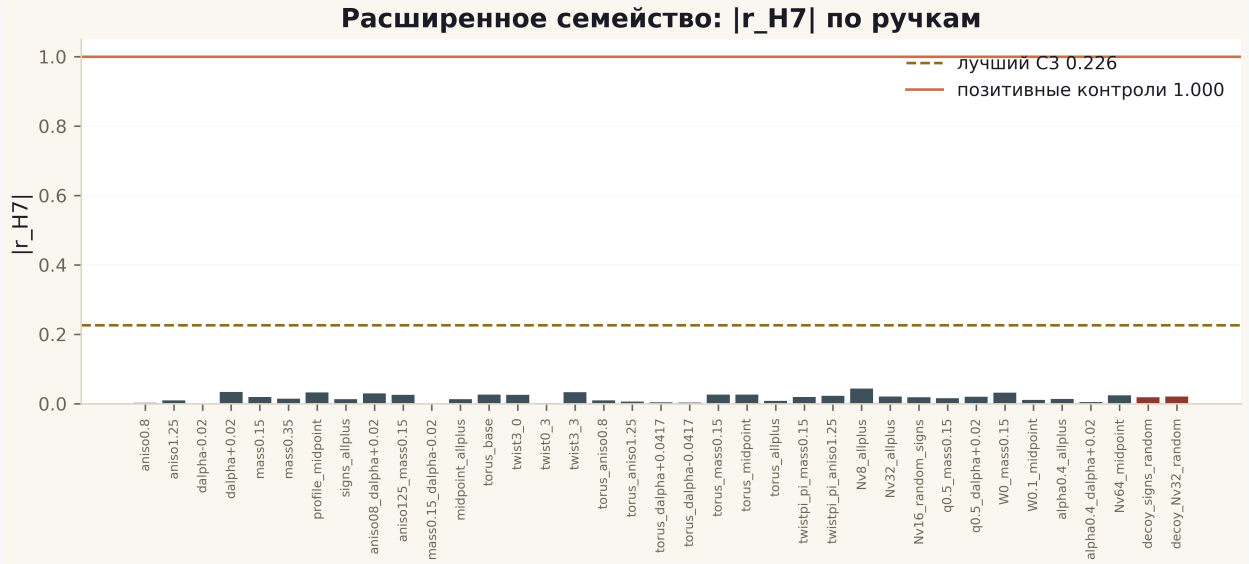


Fig. 1. |r_H7| across the 37 extended gates: gold line — best protocol value 0.226, copper — positive controls 1.000; red bars — decoys.

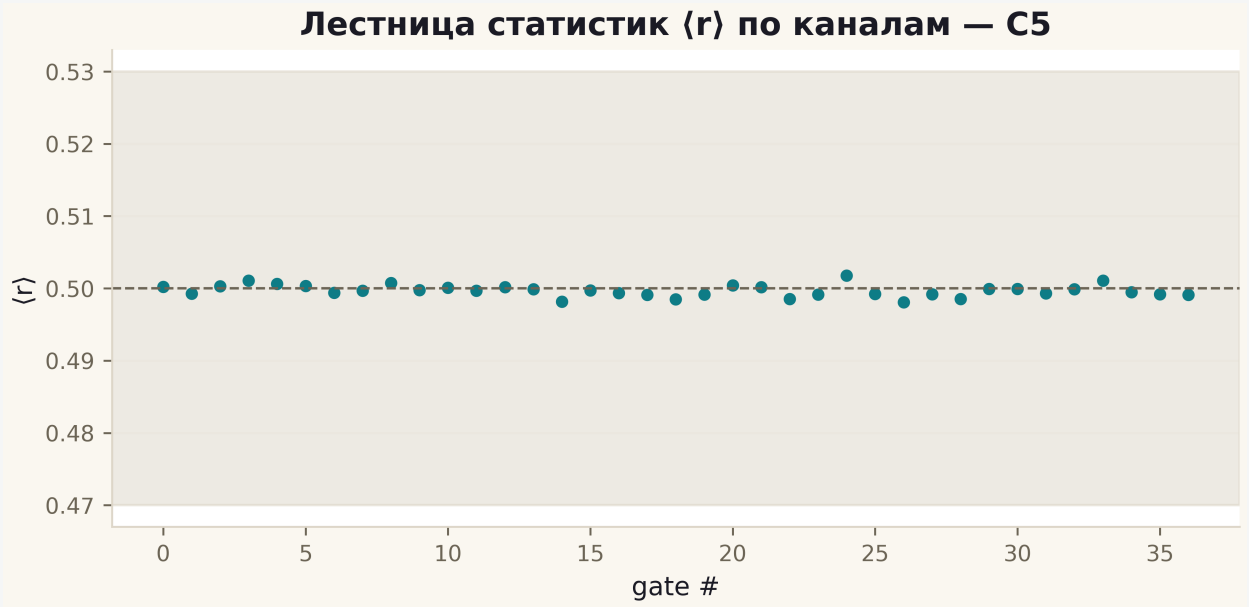


Fig. 2. ⟨r⟩ across the extended family: every point in the 0.47–0.53 band — the GUE class is universal for all handles.

Conclusions

The H2 gap is stable against a six-dimensional extension of the handle space. Combined with C3 this means: in the class of local Hermitian Hamiltonians with vortex phases, GUE universality is an attractor that erases the arithmetic setting no matter how it is imposed. A hypothetical operator carrying the $\Lambda(n)/\sqrt{n}$ weight in its trace (T6) cannot be obtained by local deformation of the lattice resonator — a statement now supported by

87 configurations across two campaigns.